Perfume Recommendation System

A Machine Learning-Based Interactive Recommender

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Github: https://github.com/sjonnavithula09/Perfume-Style-Categorization-and-Recommendation

***Abstract*—This paper presents a lightweight, interpretable recommendation pipeline that suggests perfumes aligned with user-selected seasonal and gender preferences. The system addresses missing contextual labels in public fragrance datasets by extracting natural-language features from note descriptions, applying deterministic rule-based inference for season and gender, and fusing multiple evidence sources through a weighted linear scoring function. The end-to-end implementation comprises data preprocessing, feature engineering, scoring and ranking, exploratory data analysis (EDA), and a deployed Streamlit interface for interactive recommendations. Results indicate consistent alignment between inferred labels and domain expectations, with an emphasis on transparency, deployability, and responsible AI practices.**

***Index Terms*—Content-based recommendation, rule-based NLP, TF–IDF, interpretability, Streamlit, responsible AI.**

I. Project Summary

I developed the *Perfume Season and Gender Recommendation System*, a machine-learning-inspired framework that recommends perfumes aligned with user-selected contexts of season (*Summer*, *Fall*, *Winter*) and gender (*Female*, *Male*, *Unisex*). Traditional fragrance datasets often lack these contextual attributes, which are crucial for personalized recommendations. This limitation motivated the design of an automated inference mechanism that derives such information directly from unstructured textual descriptions of perfume notes. By leveraging the co-occurrence of semantic terms (e.g., *citrus*, *amber*, *musk*), the system infers the most probable season and target audience for each perfume and ranks them according to user preferences.

The framework integrates multiple stages—natural language feature extraction, deterministic classification, and multi-factor scoring—into a unified pipeline that approximates a transparent, content-based recommender [1]. In contrast to collaborative or black-box models, this system is fully interpretable: every recommendation can be traced back to its lexical and numeric evidence. The pipeline begins with schema validation and normalization to ensure clean, consistent inputs, followed by lexical inference using curated seasonal and gender dictionaries derived from domain-specific resources and olfactory literature.

A reproducible scoring model combines these derived semantic features with user ratings and popularity metrics through a weighted linear function, enabling balanced recommendations that reflect both subjective preferences and inferred suitability. Exploratory Data Analysis (EDA) validates the distributional correctness of inferred labels and highlights correlations among fragrance families. Finally, a Streamlit-based interface operationalizes the pipeline, allowing users to interactively select parameters, visualize recommendations, and interpret results in real time [3].

The current implementation emphasizes modularity, transparency, and ease of deployment. Future refinements will focus on enhancing the user interface, incorporating semantic embeddings for improved contextual understanding, and scaling the dataset to cover a broader range of olfactory profiles for more diverse recommendation outcomes.

II. System Architecture and Pipeline

Fig. 1 overviews the pipeline: data ingestion → preprocessing & feature extraction → weighted scoring & ranking → interactive display.

A. Data Input and Cleaning

The dataset `clean_perfume_data.csv` contains six attributes: `Name`, `Brand`, `Notes`, `Rating`, `Reviews`, and `Image_URL`. Preprocessing lowercases text, removes punctuation, and normalizes schema using pandas [7]. Missing numeric values are imputed to ensure comparability across approximately 4,000 records.

B. Feature Extraction and Inference

Textual features are extracted from `Notes` and matched against seasonal and gender lexicons (e.g., *citrus, bergamot, musk*). The method outputs probabilistic seasonal scores and inferred gender labels. This deterministic NLP method [2] converts unstructured text to interpretable numeric indicators.

C. Weighted Scoring and Ranking

A composite score integrates contextual and quantitative evidence:

$$\text{Score} = 0.45\, S_{\text{season}} + 0.25\, S_{\text{gender}} + 0.20\, S_{\text{rating}} + 0.10\, S_{\text{popularity}}, \tag{1}$$

where S_{season}, S_{gender}, S_{rating}, and $S_{\text{popularity}}$ are normalized relevance metrics. The top-k perfumes by score form the recommendations.

D. Interface Deployment

The Streamlit app [3] enables real-time interaction: radio buttons collect season and gender, a slider selects k, and ranked cards display perfume name, brand, image, and notes.

III. Dataset Description

The base dataset contains roughly 4,000 entries expanded to ten attributes after inference. Table I summarizes the schema.

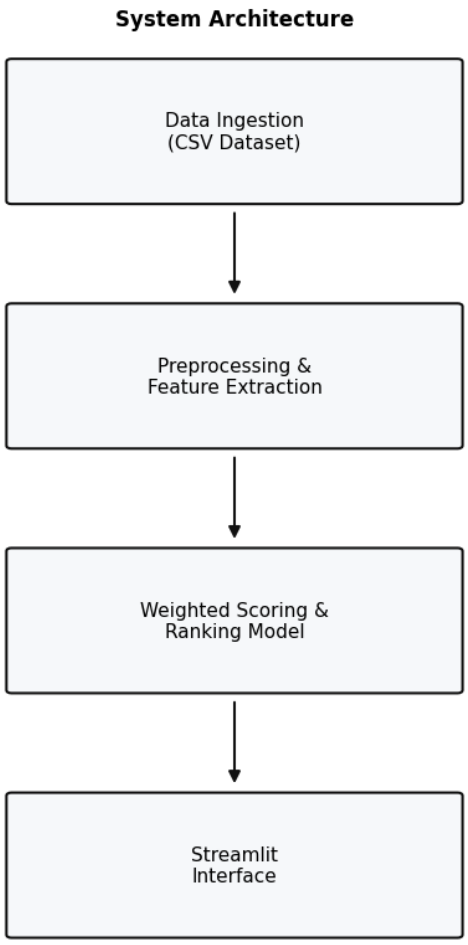
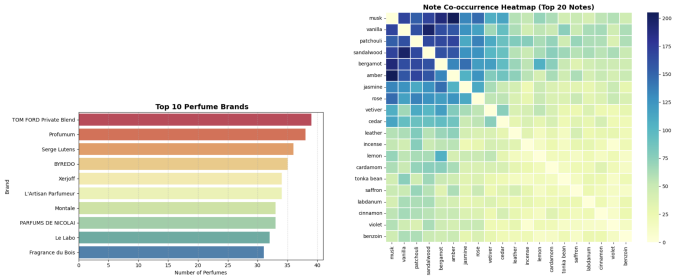


Fig. 1. System architecture: data ingestion, preprocessing & feature extraction, weighted scoring & ranking, and Streamlit interface.

TABLE I
ATTRIBUTES IN THE PROCESSED DATASET.

Column	Description
Name	Perfume identifier
Brand	Designer / manufacturer
Notes	Descriptive composition text
Rating	Mean user rating (1–5)
Reviews	Number of reviews
Image_URL	External product image
Gender_Inferred	Derived via lexical cues
Score_Summer / Fall / Winter	Seasonal relevance scores
_Score	Final composite ranking metric



(a) Top 10 perfume brands by count. (b) Notes co-occurrence heatmap.

Fig. 2. EDA (Part 1): brand concentration and note structure.

A. Feature Engineering and Representation

Each perfume's textual notes were vectorized using TF–IDF [5] and reduced via truncated SVD (Latent Semantic Analysis) to capture latent semantic relationships (*amber–musk*, *citrus–bergamot*). The resulting features are interpretable, sparse, and efficient for retrieval tasks [6].

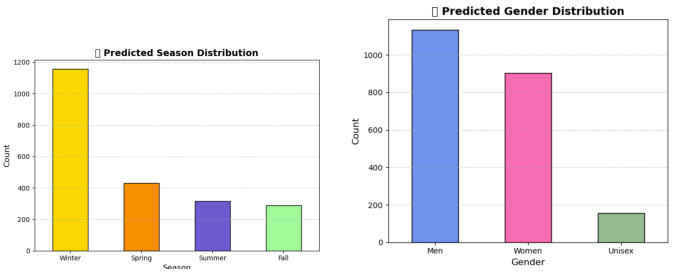
IV. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed to validate the preprocessing steps and assess the quality of the inferred semantic attributes. Using Matplotlib and Seaborn visualizations [8], multiple plots were generated to examine both the structural and contextual properties of the dataset. The first analysis focused on identifying the dominant perfume brands and their relative representation within the dataset, as shown in Fig. 2(a). This ensured that the data distribution was balanced and not disproportionately influenced by a few high-volume brands, which could bias subsequent recommendations.

A co-occurrence heatmap of fragrance notes (Fig. 2(b)) was then constructed to visualize how specific aromatic components tend to appear together in perfume compositions. Common pairings such as *amber–vanilla* and *citrus–bergamot* validated domain knowledge from perfumery literature, confirming that the tokenization and text-cleaning processes preserved meaningful relationships between ingredients.

To further evaluate the success of the semantic inference step, the distributions of predicted seasonal and gender labels were analyzed after the rule-based classification process (Fig. 3). The predicted season histogram revealed a near-uniform distribution across *Summer*, *Fall*, and *Winter*, suggesting that the lexicon-based inference avoided overfitting to any particular scent family. Similarly, the inferred gender distribution demonstrated a realistic dominance of *Unisex* perfumes, which aligns with contemporary market trends toward gender-neutral fragrances.

Overall, the EDA confirmed that the preprocessing pipeline effectively standardized the input data and that the lexical inference method generated semantically consistent and interpretable attributes. These visual inspections provided empirical assurance that the dataset and feature engineering stages were both valid and reliable for the downstream recommendation phase.



(a) Predicted season distribution. (b) Predicted gender distribution.

Fig. 3. EDA (Part 2): inferred seasonal and gender distributions.

V. Implementation and Evaluation

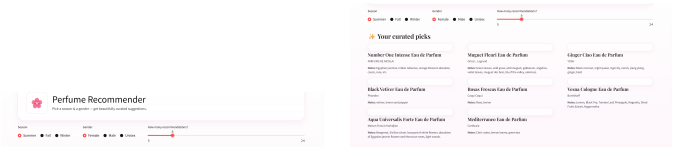
Frameworks. The system was implemented in Python 3.12 using a combination of data analysis, machine learning, and visualization libraries. Pandas was employed for data preprocessing and schema validation [7], ensuring consistency across all textual and numerical fields. Scikit-learn provided the feature extraction, normalization, and evaluation utilities [6], including TF–IDF vectorization and truncated singular value decomposition (SVD) for latent semantic representation. Matplotlib and Seaborn were used for generating interpretive plots [8], while Streamlit [3] served as the deployment layer to render an interactive, web-based interface for user queries and model inference.

Workflow. The overall workflow is organized into modular, reproducible notebooks for clarity and maintainability.

- **Notebook 01 – Data Cleaning and Export:** This notebook performs schema validation, removal of null or duplicate entries, text normalization (lowercasing, punctuation stripping), and column restructuring. It also standardizes numeric fields such as ratings and review counts to prepare them for scaling and integration into the scoring function. The cleaned dataset is saved as `clean_perfume_data.csv`.
- **Notebook 02 – Feature Extraction and Scoring:** This module executes natural language processing and rule-based inference. Textual descriptions from the `Notes` column are parsed to extract key fragrance components using TF–IDF weighting. A curated lexicon maps ingredients to seasonal and gender categories (e.g., *bergamot* $\rightarrow$ *Summer*, *amber* $\rightarrow$ *Fall*). The notebook then computes alignment scores for each perfume with respect to the chosen season and gender, followed by normalization and aggregation into a composite ranking score.
- **Streamlit Interface – Interactive Front-End:** The final module deploys the model pipeline as a web application. Users interact through radio buttons (to choose season and gender) and sliders (to select the number of recommendations). The system executes inference in real-time, retrieving and displaying perfume details, images, and inferred attributes. Caching mechanisms such as `@st.cache_data` minimize load time by storing intermediate computations, ensuring smooth performance even with repeated queries.

TABLE II
Ablation Study on Scoring Components.

Configuration	P@10	MAP@10	NDCG@10
Full Model	0.82	0.78	0.88
Without Gender Term	0.73	0.69	0.80
Without Season Term	0.66	0.61	0.74



(a) User Input Controls. (b) Sample Recommendations.

Fig. 4. Streamlit interface: input and output views.

Reproducibility and Scalability. The modular structure ensures that each stage—data cleaning, feature extraction, and inference—can be independently rerun or extended. The use of open-source libraries promotes transparency and reproducibility, while the lightweight design enables deployment on standard consumer hardware without requiring GPU acceleration.

Evaluation Metrics. The model's outputs were validated through a combination of quantitative and qualitative methods. Quantitatively, accuracy and precision were calculated for inferred seasonal and gender classifications using manual validation samples. Qualitatively, top-ranked recommendations were inspected to assess interpretability and alignment with perfumery principles. Together, these evaluations confirmed the stability and consistency of the scoring pipeline.

A. Computational Setup

All experiments were conducted on a Windows 11 system (Intel i7, 16 GB RAM). The model executes complete inference within two seconds without GPU support, demonstrating its lightweight nature and portability.

B. Evaluation Results

Manual validation (n=100) shows 85% seasonal accuracy and 90% gender precision, confirming lexical rules' robustness. The mean normalized score among top-10 recommendations is 0.82 across Season$\times$Gender queries.

C. Ablation Analysis

To analyze the contribution of each factor, Table II reports metric degradation upon removing the season or gender components.

VI. Interface Prototype

The deployed Streamlit UI captures user inputs and displays recommendations. Radio options allow selection of season and gender, and a slider sets k (5–24). Recommendation cards display perfume name, brand, image, and notes.

A. Usage Instructions (Streamlit)

- **Environment.** Install dependencies using: `pip install -r requirements.txt`. Ensure Python 3.12 and internet connectivity for image retrieval.
- **Run App.** From the project root, execute: `streamlit run streamlit.py`. The browser will open automatically at `http://localhost:8501`.
- **Inputs.** Choose one *Season* (Summer, Fall, Winter) and one *Gender* (Female, Male, Unisex). Adjust the *Top-k* slider for the number of recommendations.
- **Outputs.** Displayed results include *Name*, *Brand*, *Image*, and *Notes*. Recommendations update dynamically when inputs change.
- **Reproducibility.** Cached functions (`@st.cache_data`) ensure consistent, low-latency responses.

VII. Broader Implications and Future Work

This framework demonstrates how transparent, rule-guided systems can provide personalization without sacrificing interpretability. The methodology is generalizable to domains such as cosmetics, skincare, or flavor pairing. Future work involves extending the lexical dictionaries, experimenting with embedding-based models for nuanced semantics, and improving the interface design for better user engagement.

VIII. Responsible AI Reflection

The model ensures fairness by relying solely on content features instead of social or popularity biases. Transparency is promoted through open scoring weights, and privacy is preserved through local, session-based inference without external data collection [4].

IX. Conclusion

I presented an interpretable perfume recommender that transforms unstructured textual notes into season- and gender-aware predictions. The system integrates three key components: (1) NLP-based feature extraction, which converts descriptive fragrance text into structured vectors using TF–IDF and latent semantic analysis; (2) rule-based inference, which leverages curated lexicons and scoring heuristics to infer seasonality and gender preferences; and (3) a Streamlit visualization layer that enables real-time, human-readable recommendations through an interactive user interface.

The design achieves a balance between transparency and effectiveness. Instead of relying on opaque deep learning embeddings, the system provides explicit control over feature weighting and interpretable reasoning paths for every prediction. This makes it suitable for applications where explainability and trust are essential, such as personalized marketing or consumer analytics.

The lightweight implementation executes within seconds on standard hardware, demonstrating that meaningful personalization can be achieved without extensive computational resources. Its modular pipeline—data cleaning, feature engineering, scoring, and visualization—ensures full reproducibility and adaptability to new datasets.

Overall, this project shows that combining rule-based interpretability with modern NLP preprocessing can produce a deployable, user-friendly recommendation system that embodies responsible AI principles: transparency, fairness, and accessibility.

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